Methods for Handling Multiple Outcomes in Health Data

Sivakamakshi Muthu Kumarasamy   Shivadarshini Sreekanth

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Supervisor: Autumn Johnson

Department: Department of Mathematics and Statistical Sciences

Affiliation: University of Galway

Location: Galway, Ireland

Emails: s.muthukumarasamy1@universityofgalway.ie; s.sreekanth1@universityofgalway.ie

1 Introduction

1.1 Background

Clinical research frequently involves more than one outcome of interest. In oncology and chronic disease studies, patients may experience several clinically important events during follow-up, including disease progression, recurrence, metastasis, hospitalisation, and death. These outcomes are often analysed separately using conventional methods, but separate analyses may not fully reflect the patient journey because events can be clinically and statistically related (1–3).

Time-to-event outcomes are especially important in cancer research because not all patients experience the event during the observed follow-up period. Patients who do not experience the event are censored, meaning that their event time is only partially observed. Survival methods such as Kaplan–Meier estimation, log-rank tests, and Cox proportional hazards regression are therefore widely used in clinical trials (4–7). However, these methods are usually applied to one endpoint at a time.

1.2 Multiple Outcomes in Clinical Research

Multiple outcomes matter because they may represent different levels of clinical importance. For example, death is usually more serious than disease progression, while progression may still be important because it can reduce quality of life, lead to treatment switching, and increase the future risk of death. A statistical method for multiple outcomes should ideally account for timing, censoring, clinical hierarchy, and the dependency between outcomes (8–10).

Composite endpoints are often used to combine several events into a single endpoint. A participant is considered to have the composite event if any component occurs. This can increase the number of events and statistical power, but it may also be misleading when components differ substantially in clinical importance (2,3,11). For example, a composite endpoint combining progression and death may be driven mainly by progression events, even though death is the more serious outcome.

1.3 Challenges in Analysing Multiple Outcomes

Analysing multiple outcomes creates several methodological challenges. First, repeated testing of separate endpoints can increase the probability of false positive findings. Second, outcomes may be dependent; for example, disease progression may alter the subsequent risk of death. Third, treatment effects may differ across endpoints or change over time. Fourth, proportional hazards assumptions may not always hold in survival models, especially when survival curves cross (12–14).

Advanced methods have been developed to address these challenges. The win ratio compares patients using a pre-specified hierarchy of outcomes, giving priority to the most clinically important event (8). Net-benefit approaches provide decision-focused summaries of clinical advantage (15,16). Multi-state models describe how patients move between disease states over time and can estimate transition-specific hazards and probabilities (14,17–19).

1.4 SECOMBIT Trial Overview

This project used data from the SECOMBIT trial, an open-label, multicentre, randomised phase II clinical trial involving patients with metastatic BRAF V600-mutated melanoma. The trial evaluated treatment sequencing strategies involving targeted therapy with encorafenib plus binimetinib and immunotherapy with ipilimumab plus nivolumab (20,21).

The trial included three treatment arms. Arm A received encorafenib plus binimetinib followed by ipilimumab plus nivolumab after disease progression. Arm B received ipilimumab plus nivolumab followed by encorafenib plus binimetinib after disease progression. Arm C used a planned "sandwich" strategy: patients received encorafenib plus binimetinib for eight weeks and then switched to ipilimumab plus nivolumab until progression, followed by encorafenib plus binimetinib after progression (20,21).

1.5 Dataset Description

The dataset used in this project was obtained from an open-source version of the SECOMBIT clinical trial data made available for research and educational analysis. The dataset contained approximately 209 patients allocated to Arms A, B, and C. The key outcomes were overall survival (OS) and progression-free survival (PFS). Overall survival was defined as the time from study entry to death, while progression-free survival was defined as the time from study entry to disease progression or death. The main covariates available for analysis were treatment arm, number of metastatic sites, lactate dehydrogenase (LDH) level, tumour mutational burden (TMB), and JAK mutation status.

The dataset was suitable for this project because it contained multiple clinically important time-to-event outcomes that are related but not equivalent. It also allowed direct comparison between traditional survival methods and more advanced approaches such as win-ratio analysis, net-benefit analysis, and multi-state modelling.

1.6 Clinical Trial Inference and Predictive Modelling Context

The methods applied in this project were used primarily in a clinical trial inference context, with the goal of estimating average treatment effects across Arms A, B, and C. Because the SECOMBIT study was randomised, comparisons between treatment arms can be interpreted as treatment-sequencing effects more strongly than would be possible in a purely observational dataset.

However, several of these methods also have relevance for predictive modelling. Multi-state models are particularly suitable for prediction because they estimate the probability that a patient in a given state will remain stable, progress, or die at future time points. When patient-level covariates such as LDH, TMB, or JAK mutation status are incorporated into transition intensities, the model can be extended to produce personalised risk predictions. By contrast, the win ratio and net benefit are mainly trial-level summary methods; they compare treatment strategies across groups but do not naturally produce individual patient predictions.

1.7 Aim and Objectives

The aim of this project was to compare statistical strategies for handling multiple outcomes in health data, using the SECOMBIT dataset as an applied case study.

The objectives were:

1. To analyse OS and PFS separately using logistic regression and survival analysis.
2. To compare individual endpoint analysis with composite endpoint analysis.
3. To evaluate treatment-arm differences using Cox proportional hazards models.
4. To apply hierarchical and decision-focused approaches using win-ratio and net-benefit methods.
5. To model disease pathways using a multi-state model.
6. To compare the strengths, limitations, and clinical interpretability of each method.

2 Methods

2.1 Data Preparation

The SECOMBIT dataset was imported into R and cleaned before analysis. Variables were recoded to create standard event-time and event-status variables for OS, PFS, and the composite endpoint. Treatment arm and clinical covariates were coded as categorical variables. A composite endpoint was created by defining an event when either progression or death occurred.

2.2 Missing Data Handling

Missing data were assessed before fitting adjusted models. Missing values were present mainly in the JAK mutation variable. No formal imputation was performed. Therefore, regression models requiring all covariates used complete-case analysis. This means that participants with missing values in any covariate included in the model were excluded from that specific analysis.

Complete-case analysis is simple and transparent, but it can reduce statistical power and may introduce bias if data are not missing completely at random (22,23). This limitation is especially relevant for multivariable Cox and logistic regression models, where the inclusion of JAK mutation status reduced the effective sample size.

Table 1 below summarises the number of missing values for each variable in the dataset. Notably, TMB and JAK mutation each had 126 missing values, representing approximately 60% of the total sample.

Table 1: Missing values by variable

Variable	Missing values
id	0
Arm	0
Sites	2
LDH	5
TMB	126
JAK	126
pfs_time	3
pfs_status	3
os_time	3
os_status	3
comp_time	3
comp_status	3
binary_os	3
binary_pfs	3
binary_comp	3

2.3 Statistical Methods

2.3.1 Logistic Regression

Logistic regression was used as a preliminary event-based analysis for OS, PFS, and the composite endpoint. For a binary event indicator, the model can be written as:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip},$$

where p_i is the probability that patient i experiences the event and $X_{i1}, \ldots, X_{ip}$ are covariates. Odds ratios greater than 1 indicate higher odds of the event, while odds ratios less than 1 indicate lower odds. Logistic regression does not account for time-to-event information or censoring, so it was not treated as the primary method for survival outcomes (24).

2.3.2 Kaplan–Meier Estimation

Kaplan–Meier estimation was used to estimate survival probabilities over time (4). The Kaplan–Meier estimator is:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right),$$

where d_i is the number of events at time t_i and n_i is the number of individuals at risk just before t_i. Kaplan–Meier curves were generated for OS, PFS, and the composite endpoint by treatment arm.

2.3.3 Log-Rank Test

The log-rank test was used to compare survival curves across treatment arms (5). It tests the null hypothesis that there is no difference in survival experience between groups. The test is sensitive to differences in hazards over time, but it does not directly estimate the size or direction of the treatment effect.

2.3.4 Cox Proportional Hazards Model

Cox proportional hazards models were used to estimate adjusted treatment effects for OS, PFS, and the composite endpoint (6). The model is:

$$h(t|X) = h_0(t)\exp(\beta X),$$

where $h(t|X)$ is the hazard at time t for covariate vector X, $h_0(t)$ is the baseline hazard, and β represents the log hazard ratio. Hazard ratios greater than 1 indicate increased instantaneous risk, while hazard ratios less than 1 indicate reduced instantaneous risk.

The key assumption is proportional hazards, meaning that hazard ratios remain constant over follow-up time. This assumption was assessed using Schoenfeld residuals (12,25).

2.3.5 Composite Endpoint Analysis

The composite endpoint was defined as the first occurrence of progression or death. Composite endpoints can increase the number of events and simplify interpretation, but they may be problematic when the components differ in clinical importance (2,3,11). In this project, the composite endpoint was interpreted carefully because progression and death are not clinically equivalent.

2.3.6 Win Ratio

The win ratio is a hierarchical method for comparing treatment groups using multiple outcomes (8). Patients are compared pairwise. The most clinically important outcome is evaluated first; if a pair is tied, the next outcome is considered. In this project, OS was prioritised first and PFS second.

The win ratio is defined as:

$$WR = \frac{\text{Number of wins}}{\text{Number of losses}}.$$

A win ratio greater than 1 favours the treatment group, while a value less than 1 favours the comparator group. This method is clinically meaningful because it respects the hierarchy of outcomes.

2.3.7 Net Benefit

Net benefit was estimated using generalised pairwise comparisons through the `BuyseTest` package. Net benefit can be expressed as:

$$NB = P(\text{Win}) - P(\text{Loss}),$$

where $P(\text{Win})$ is the probability that a patient in one group has a better outcome than a comparable patient in another group, and $P(\text{Loss})$ is the probability of a worse outcome (16,26). Positive net benefit indicates an overall clinical advantage.

2.3.8 Multi-State Models

Multi-state models were used to model transitions between disease states over time (14,17,18). The states were stable disease, progression, and death. The transition intensity from state r to state s is:

$$q_{rs} = \lim_{\Delta t \to 0} \frac{P(X(t + \Delta t) = s | X(t) = r)}{\Delta t}.$$

The model estimated transition-specific hazards and state occupation probabilities. A Markov assumption was used, meaning that future transitions depend on the current state and covariates rather than the full previous history.

2.4 State-Transition Structure

In multi-state modelling, it is important to distinguish between the theoretical clinical pathway and the pathway actually observed in the dataset. The full theoretical model allows patients to move from stable disease to progression, from progression to death, and directly from stable disease to death. This direct stable-to-death transition is clinically possible because some patients may die before documented progression.

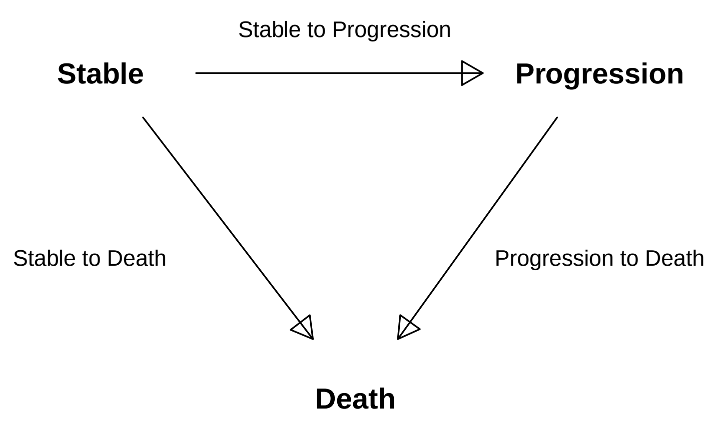

Figure 1: Full theoretical state-transition diagram for the multi-state model

In the SECOMBIT dataset used here, no patient died before progression. Therefore, the observed model fitted in this analysis used the simplified pathway shown below, with transitions from stable disease to progression and from progression to death only.

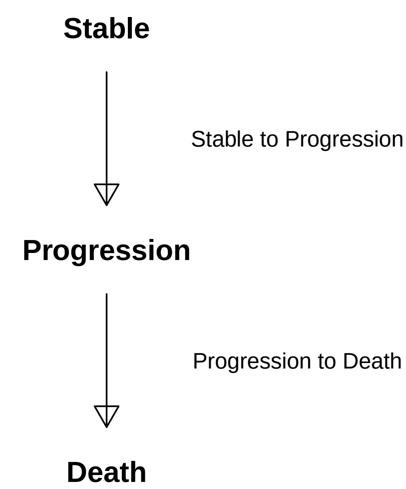

Figure 2: Observed state-transition diagram fitted in this analysis

2.5 Software Implementation

All analyses were conducted in R version 4.5.3 (2026-03-11 ucrt). The main packages used were `survival` for survival modelling, `survminer` for Kaplan–Meier plots, `msm` for multi-state modelling, `WinRatio` for win-ratio analysis, `BuyseTest` for net-benefit analysis, `tableone` for descriptive baseline tables, `broom` for model summaries, and `knitr`/`kableExtra` for table formatting (17,27–30).

3 Results

3.1 Descriptive Summary

Table 2 below provides descriptive statistics for OS and PFS in months. The mean OS and PFS values describe the overall follow-up pattern in the study sample.

Table 2: Numeric summary of OS and PFS in months

Statistic	OS	PFS
mean	32.39	28.81
sd	17.19	18.06
median	35.90	29.80
min	1.13	1.10
max	65.50	65.50

3.2 Baseline Characteristics

Only variables measured at or before treatment initiation were included in the baseline table.

Table 3 below presents baseline characteristics by treatment arm. This helps assess whether measured prognostic factors were broadly balanced across Arms A, B, and C before outcome analysis.

Table 3: Baseline characteristics by treatment arm

	level	A	B	C	p	test
n		69	71	69		
Sites (%)	1 - 2	43 (62.3)	41 (58.6)	42 (61.8)	0.887	
	>= 3	26 (37.7)	29 (41.4)	26 (38.2)		
LDH (%)	Normal	41 (61.2)	41 (59.4)	48 (70.6)	0.346	
	Elevated	26 (38.8)	28 (40.6)	20 (29.4)		
TMB (%)	< 10	20 (71.4)	17 (68.0)	18 (60.0)	0.639	
	>= 10	8 (28.6)	8 (32.0)	12 (40.0)		
JAK (%)	Wild Type	24 (82.8)	17 (70.8)	16 (53.3)	0.050	
	Mutated	5 (17.2)	7 (29.2)	14 (46.7)		

3.3 Event Counts by Treatment Arm

Table 4 and Figure 3 below show the number of OS, PFS, and composite events in each treatment arm. These counts provide an initial descriptive comparison before model-based analysis.

Table 4: Event counts by treatment arm

Arm	n	OS_Events	PFS_Events	Composite_Events
A	69	35	49	49
B	71	23	30	30
C	69	27	32	32

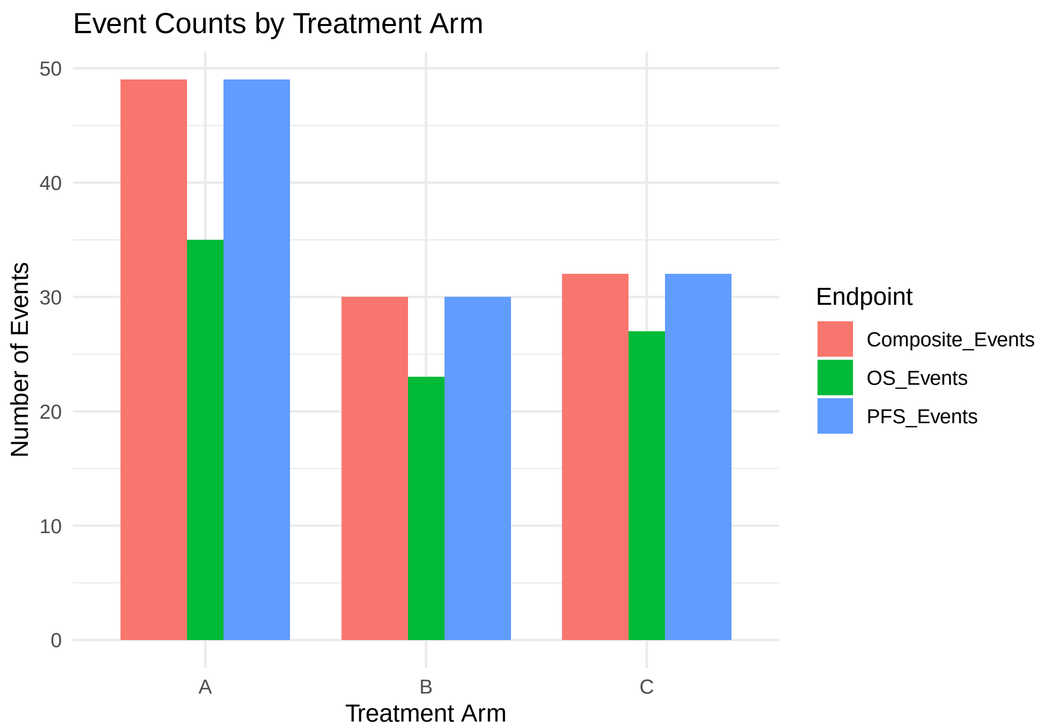

Figure 3: OS, PFS, and composite event counts by treatment arm

3.4 Logistic Regression Results

Tables 5, 6, and 7 below present the adjusted logistic regression results for the OS, PFS, and composite endpoints respectively. These models estimate odds ratios for experiencing each event, but do not account for censoring or event timing.

Table 5: Logistic regression for OS event

Variable	OR (95% CI)	p-value
(Intercept)	0.64 (0.25–1.61)	0.3490
ArmB	0.70 (0.20–2.34)	0.5700
ArmC	0.83 (0.25–2.70)	0.7500
Sites>= 3	2.45 (0.88–7.13)	0.0908
LDHElevated	0.79 (0.26–2.27)	0.6650
TMB>= 10	0.71 (0.23–2.03)	0.5230
JAKMutated	0.52 (0.15–1.64)	0.2730

Table 6: Logistic regression for PFS event

Variable	OR (95% CI)	p-value
(Intercept)	1.33 (0.53–3.42)	0.5450
ArmB	0.43 (0.12–1.44)	0.1800
ArmC	1.08 (0.34–3.59)	0.8940
Sites>= 3	1.83 (0.65–5.37)	0.2590
LDHElevated	0.90 (0.30–2.62)	0.8410

Variable	OR (95% CI)	p-value
TMB>= 10	0.72 (0.25–2.07)	0.5460
JAKMutated	0.26 (0.07–0.80)	0.0247

Table 7: Logistic regression for composite event

Variable	OR (95% CI)	p-value
(Intercept)	1.33 (0.53–3.42)	0.5450
ArmB	0.43 (0.12–1.44)	0.1800
ArmC	1.08 (0.34–3.59)	0.8940
Sites>= 3	1.83 (0.65–5.37)	0.2590
LDHElevated	0.90 (0.30–2.62)	0.8410
TMB>= 10	0.72 (0.25–2.07)	0.5460
JAKMutated	0.26 (0.07–0.80)	0.0247

OS event (Table 5). No covariate reached statistical significance. Arm B (OR = 0.70, p = 0.570) and Arm C (OR = 0.83, p = 0.750) both showed odds ratios below 1 relative to Arm A, but with wide confidence intervals crossing 1. Having three or more metastatic sites showed the strongest trend (OR = 2.45, p = 0.0908), suggesting higher odds of death with greater disease burden, though this fell short of statistical significance.

PFS event (Table 6). Arm B again showed lower odds of a PFS event relative to Arm A (OR = 0.43, p = 0.180), while Arm C was close to 1 (OR = 1.08, p = 0.894); neither treatment-arm comparison was statistically significant. JAK mutation status was the only significant covariate: mutated JAK was associated with substantially lower odds of progression (OR = 0.26, 95% CI 0.07–0.80, p = 0.0247), although this was based on the smaller complete-case sample.

Composite event (Table 7). Results were identical to the PFS model, including the same JAK effect (OR = 0.26, p = 0.0247). This is not an error: because the composite endpoint combines progression and death, and no patient died before progressing in this dataset, every PFS event was also a composite event. The two models were therefore mathematically equivalent here, a pattern that recurs later in the Kaplan–Meier and Cox results (Sections 3.5–3.6).

Overall, treatment arm showed no statistically significant association with event odds in any logistic regression model, although Arm B consistently trended lower than Arm A for both OS and PFS. JAK mutation was significant only for PFS and the composite endpoint, not OS, suggesting a progression-specific rather than survival-specific association. However, logistic regression cannot confirm this distinction because it ignores event timing.

This points to the core limitation of the approach: logistic regression collapses each outcome into a binary event or no-event indicator by the end of follow-up, discarding information about when the event occurred and treating censored patients as equivalent regardless of how long they were followed. Logistic regression was therefore useful only as a preliminary event-based analysis, while survival models were more appropriate for the main OS and PFS analyses.

3.5 Kaplan–Meier Survival Curves and Log-Rank Tests

Figures 3, 4, and 5 below show the Kaplan–Meier survival curves by treatment arm for OS, PFS, and the composite endpoint respectively. The corresponding log-rank p-values are displayed on each figure.

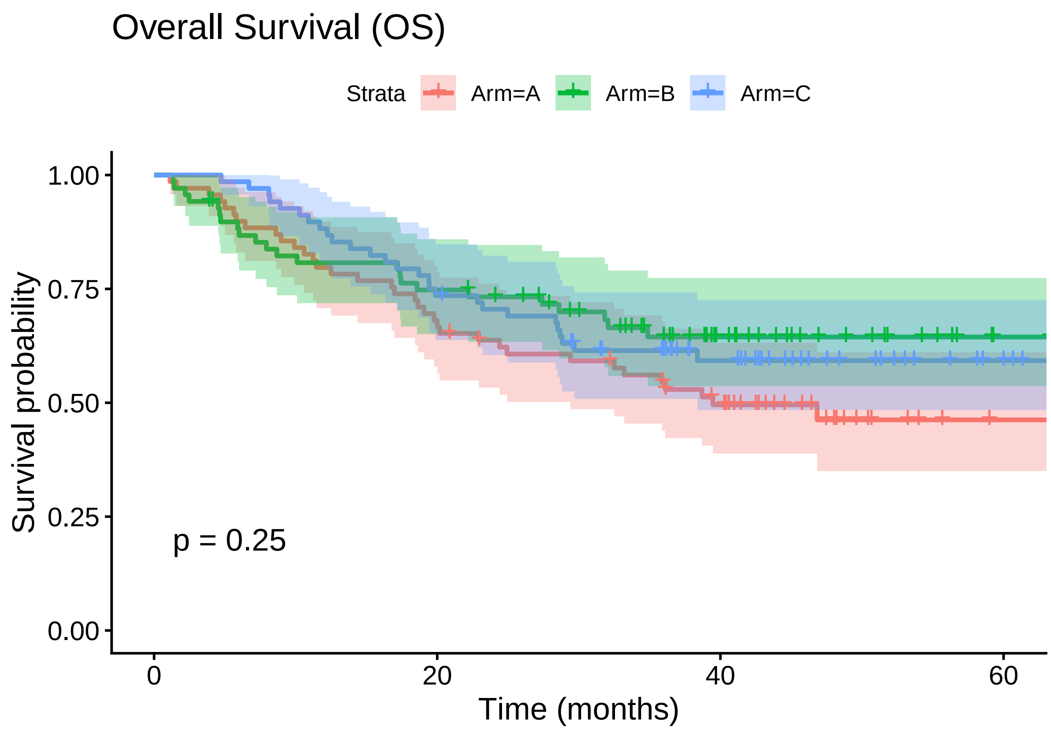

Figure 4: Kaplan–Meier curve for overall survival by treatment arm

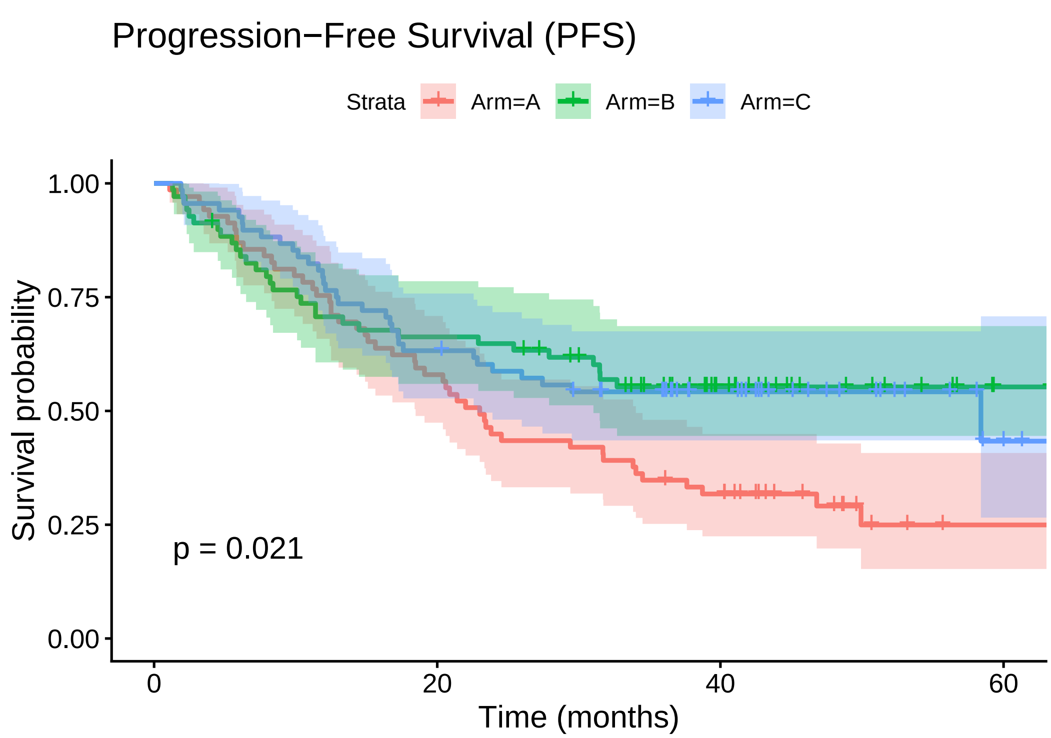

Figure 5: Kaplan–Meier curve for progression-free survival by treatment arm

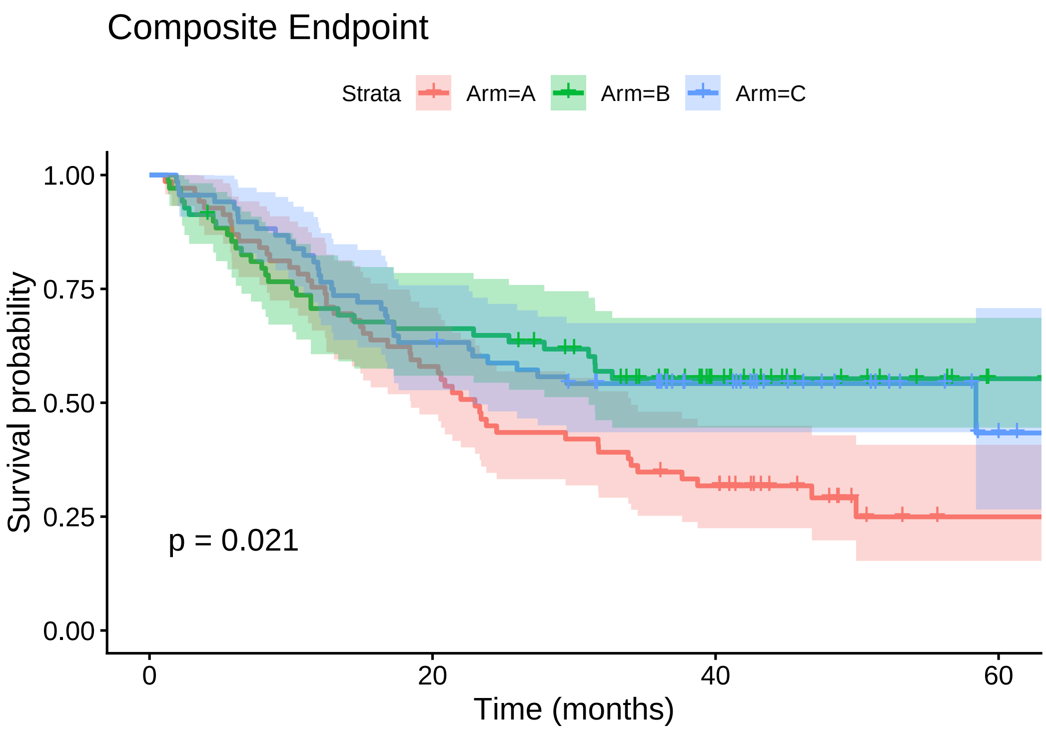

Figure 6: Kaplan–Meier curve for composite endpoint by treatment arm

Table 8 below summarises the log-rank test statistics and p-values for each endpoint. This provides a formal unadjusted comparison of survival curves across treatment arms.

Table 8: Log-rank test results

Endpoint	Chi.square	df	p.value
OS	2.80	2	0.246
PFS	7.74	2	0.021
Composite Endpoint	7.74	2	0.021

The Kaplan–Meier curves for OS appeared relatively close, and the OS log-rank p-value was approximately 0.25, suggesting no statistically significant difference in overall survival across arms. The PFS curves separated more clearly, with a log-rank p-value near 0.02, indicating evidence that at least one arm differed in progression-free survival.

The composite endpoint produced results that were effectively identical to PFS, with the same rounded log-rank p-value and visually indistinguishable Kaplan–Meier curves (Figures 5 and 6). This is not a coincidence or an error. The composite endpoint was defined as the first occurrence of progression or death, and since no patient in this dataset died before progressing, every progression event was also recorded as the composite event, and no patient contributed a composite event that was not already a PFS event. As a result, the composite endpoint variable was numerically equivalent to the PFS variable for every patient, and any analysis applied to one necessarily reproduced the other. This finding is consistent with the equivalence already noted in the logistic regression results (Section 3.4) and recurs in the Cox model results (Section 3.6). It also highlights an important limitation of composite endpoints in general: when one component event, such as progression, occurs far more frequently than death, the composite endpoint is effectively driven by the more common event and may add no additional information beyond that endpoint alone, despite combining two outcomes of clearly different clinical importance.

3.6 Cox Proportional Hazards Models

Tables 9, 10, and 11 below present the adjusted Cox proportional hazards model results for OS, PFS, and the composite endpoint respectively. Each model adjusts for treatment arm, number of metastatic sites, LDH level, TMB status, and JAK mutation status using complete-case analysis.

Table 9: Cox model for OS

Variable	HR (95% CI)	p-value
ArmB	0.98 (0.38–2.50)	0.9590
ArmC	0.83 (0.33–2.07)	0.6900
Sites>= 3	2.02 (0.93–4.42)	0.0764
LDHElevated	0.96 (0.43–2.15)	0.9130
TMB>= 10	0.72 (0.31–1.67)	0.4480
JAKMutated	0.63 (0.24–1.63)	0.3380

Table 10: Cox model for PFS

Variable	HR (95% CI)	p-value
ArmB	0.73 (0.30–1.77)	0.4870
ArmC	1.03 (0.48–2.22)	0.9470
Sites>= 3	1.60 (0.80–3.20)	0.1870
LDHElevated	1.14 (0.56–2.34)	0.7150
TMB>= 10	0.85 (0.40–1.78)	0.6660
JAKMutated	0.41 (0.16–1.03)	0.0571

Table 11: Cox model for composite endpoint

Variable	HR (95% CI)	p-value
ArmB	0.73 (0.30–1.77)	0.4870
ArmC	1.03 (0.48–2.22)	0.9470
Sites>= 3	1.60 (0.80–3.20)	0.1870
LDHElevated	1.14 (0.56–2.34)	0.7150
TMB>= 10	0.85 (0.40–1.78)	0.6660
JAKMutated	0.41 (0.16–1.03)	0.0571

In the Cox model for OS, the hazard ratios for Arm B and Arm C compared with Arm A were close to 1 with wide confidence intervals and non-significant p-values, suggesting no clear overall survival advantage in the adjusted model.

For PFS, the direction of the hazard ratios suggested several non-significant trends. Arm B had a hazard ratio below 1 relative to Arm A (HR = 0.73, 95% CI 0.30–1.77, p = 0.487), consistent with a potentially protective effect on progression, but the wide confidence interval crossing 1 and the non-significant p-value mean this cannot be interpreted as statistically reliable evidence of a treatment effect once covariates are adjusted for.

JAK mutation showed the strongest trend in the model (HR = 0.41, 95% CI 0.16–1.03, p = 0.0571), narrowly missing the conventional 5% significance threshold. Conversely, having three or more metastatic sites showed a hazard ratio above 1 (HR = 1.60, 95% CI 0.80–3.20, p = 0.187), suggesting a possible increase in progression risk with greater baseline disease burden, while elevated LDH showed a hazard ratio close to but slightly above 1 (HR = 1.14, 95% CI 0.56–2.34, p = 0.715), indicating essentially no adjusted effect.

None of these covariates, including JAK mutation, reached statistical significance at the 5% level in the adjusted Cox model, and all associated confidence intervals included 1. These directional patterns are therefore best described as suggestive, non-significant trends rather than confirmed effects, and should be interpreted with caution given the reduced complete-case sample size used for this adjusted model.

To assess potential confounding and the effect of complete-case reduction, treatment effects were estimated using both crude and adjusted models. Crude models included treatment arm only and therefore used the full available treatment-arm sample, whereas adjusted models additionally controlled for Sites, LDH, TMB, and JAK mutation status.

Table 12: Crude and adjusted logistic regression treatment effects. Values are odds ratios with 95% confidence intervals and p-values.

Outcome	Model	Arm B vs Arm A	Arm C vs Arm A
OS	Crude	0.49 (0.24–0.96) (p = 0.0397)	0.64 (0.32–1.26) (p = 0.196)
OS	Adjusted	0.70 (0.20–2.34) (p = 0.57)	0.83 (0.25–2.70) (p = 0.75)
PFS	Crude	0.31 (0.15–0.63) (p = 0.00128)	0.36 (0.18–0.73) (p = 0.00483)
PFS	Adjusted	0.43 (0.12–1.44) (p = 0.18)	1.08 (0.34–3.59) (p = 0.894)
Composite	Crude	0.31 (0.15–0.63) (p = 0.00128)	0.36 (0.18–0.73) (p = 0.00483)
Composite	Adjusted	0.43 (0.12–1.44) (p = 0.18)	1.08 (0.34–3.59) (p = 0.894)

Table 13: Crude and adjusted Cox proportional hazards treatment effects. Values are hazard ratios with 95% confidence intervals and p-values.

Outcome	Model	Arm B vs Arm A	Arm C vs Arm A
OS	Crude	0.66 (0.39–1.12) (p = 0.125)	0.73 (0.44–1.21) (p = 0.221)
OS	Adjusted	0.98 (0.38–2.50) (p = 0.959)	0.83 (0.33–2.07) (p = 0.69)
PFS	Crude	0.58 (0.37–0.91) (p = 0.0187)	0.60 (0.39–0.94) (p = 0.026)
PFS	Adjusted	0.73 (0.30–1.77) (p = 0.487)	1.03 (0.48–2.22) (p = 0.947)
Composite	Crude	0.58 (0.37–0.91) (p = 0.0187)	0.60 (0.39–0.94) (p = 0.026)
Composite	Adjusted	0.73 (0.30–1.77) (p = 0.487)	1.03 (0.48–2.22) (p = 0.947)

Comparing crude and adjusted estimates helps assess whether treatment-arm effects are robust to covariate adjustment. Similar estimates would support balanced randomisation, while material differences suggest possible confounding or complete-case selection effects.

In this dataset, the crude estimates based on the full available sample showed significant protective effects of Arm B and Arm C on PFS and the composite endpoint. For example, the crude Cox model gave HR = 0.58 for Arm B (p = 0.0187) and HR = 0.60 for Arm C (p = 0.026). After adjustment for Sites, LDH, TMB, and JAK status using the complete-case sample, all significance was lost: Arm B's hazard ratio increased to 0.73 (p = 0.487), and Arm C's increased to 1.03 (p = 0.947).

Two factors likely explain this change. First, JAK mutation prevalence differed across arms and was itself protective against progression, so part of the crude treatment-arm difference may reflect this imbalance rather than treatment sequence alone. Second, the complete-case restriction excluded a large proportion of patients because TMB and JAK were missing for many cases, substantially widening confidence intervals and reducing power.

The consistency of this pattern across both logistic and Cox models suggests that the crude treatment effects were not fully robust to adjustment, with both measured covariate imbalance and reduced complete-case sample size likely contributing.

Table 12 below reports the global Schoenfeld residual test results for the proportional hazards assumption. These tests help determine whether a single Cox hazard ratio is appropriate over the full follow-up period.

Table 14: Proportional hazards assumption tests

Endpoint	PH.global.p.value
OS	0.0581
PFS	0.0119
Composite Endpoint	0.0119

The Schoenfeld residual tests indicated a borderline violation of the proportional hazards assumption for OS and a clearer violation for PFS and the composite endpoint. A violated proportional hazards assumption means that the hazard ratio between treatment arms is not constant over follow-up time, so reporting a single summary hazard ratio may be misleading.

This finding provides direct methodological justification for the use of the multi-state model. Rather than summarising the entire PFS trajectory with one hazard ratio, the multi-state model estimates separate transition-specific hazard ratios for the stable-to-progression and progression-to-death transitions. This decomposition allows the analysis to show whether treatment sequencing mainly affects progression, post-progression mortality, or both.

Several alternative approaches can be used when proportional hazards assumptions are violated. Restricted mean survival time (RMST) summarises the expected event-free time up to a pre-specified time horizon and does not rely on the proportional hazards assumption (31). Weighted log-rank tests, such as Fleming–Harrington tests, allow different parts of the survival curve to be emphasised and may be useful when treatment effects occur earlier or later during follow-up (32,33). Landmark analysis can also be used to compare survival beyond a fixed time point, particularly when treatment effects or prognostic factors vary over time (34). Time-varying coefficient Cox models are another option for non-proportional hazards, but they were not pursued here because this dataset did not contain repeated longitudinal measurements of covariates over follow-up, and the aim of the project was to compare standard endpoint, hierarchical, and multi-state approaches rather than to develop an extended Cox model (25).

3.7 Win-Ratio Results

Table 13 below shows the pairwise win-ratio results. A win ratio greater than 1 indicates that the first-named arm won more matched pairs than it lost when OS and PFS were evaluated hierarchically.

Table 15: Win-ratio comparison of treatment arms

Comparison	Win ratio (95% CI)	p-value
B vs A	1.49 (0.93–2.40)	0.0997
C vs A	1.59 (1.00–2.52)	0.0503
C vs B	1.06 (0.61–1.84)	0.8372

The win-ratio analysis favoured Arm B over Arm A, with a win ratio of 1.49 (95% CI 0.93–2.40; p = 0.0997). Arm C also favoured over Arm A, with a win ratio of 1.59 (95% CI 1.00–2.52; p = 0.0503). Arm C compared with Arm B was close to neutral, with a win ratio of 1.06 (95% CI 0.61–1.84; p = 0.8372). None of these comparisons reached statistical significance at the conventional 5% threshold, and all three confidence intervals included or nearly touched 1, meaning the observed win ratios cannot be interpreted as confirmed treatment effects. Overall, Arms B and C showed favourable hierarchical outcome trends compared with Arm A, but these trends did not reach statistical significance, and there was no clear superiority between Arms B and C.

3.8 Net-Benefit Results

The net-benefit analysis was used to compare treatment strategies by combining information from overall survival (OS) and progression-free survival (PFS). Table 16 summarises the overall net benefit for each treatment comparison together with the contributions from the individual endpoints. Table 17 presents the endpoint-specific estimates with their corresponding confidence intervals and p-values.

Table 16: Overall net benefit and endpoint-specific contributions by treatment comparison

Comparison	Global Net Benefit	OS Contribution	PFS Contribution
B vs A	0.220	0.134	0.086
C vs A	0.202	0.132	0.069
C vs B	-0.049	-0.010	-0.039

Table 17: Endpoint-specific net-benefit estimates with confidence intervals and p-values

	Endpoint - OS			Endpoint - PFS		
Comparison	OS Delta	OS CI (2.5%, 97.5%)	OS p-value	PFS Delta	PFS CI (2.5%, 97.5%)	PFS p-value
B vs A	0.133	-0.05; 0.31	0.15	0.220	0.02; 0.40	0.031
C vs A	0.132	-0.05; 0.31	0.15	0.201	0.0001; 0.39	0.050
C vs B	-0.010	-0.18; 0.16	0.91	-0.050	-0.25; 0.15	0.640

The net-benefit analysis supported the same broad interpretation as the win-ratio analysis. As shown in Table 16, both Arm B and Arm C demonstrated a positive overall net benefit compared with Arm A (0.220 and 0.202, respectively), whereas the comparison between Arms C and B showed little overall difference (-0.049). In both comparisons with Arm A, the contribution from overall survival (OS) was numerically greater than that from progression-free survival (PFS), indicating that improvements in OS accounted for a larger proportion of the overall clinical benefit.

Table 17 presents the endpoint-specific net-benefit estimates together with their corresponding confidence intervals and p-values. No statistical test was available for the overall global net benefit because inference was performed separately for each endpoint. The PFS component was statistically significant for both Arm B versus Arm A ($p = 0.031$) and Arm C versus Arm A ($p = 0.050$), whereas the OS component was not statistically significant for either comparison (both $p = 0.15$). There was also no significant difference between Arms C and B for either endpoint. Overall, these findings are consistent with the win-ratio analysis in Section 3.7, indicating that the observed advantage of Arms B and C over Arm A was driven primarily by improvements in progression-free survival rather than overall survival.

3.9 Multi-State Model Results

Table 15 below presents the transition-specific hazard ratios from the multi-state model by treatment arm. Arm A is the reference category, and hazard ratios below 1 indicate a reduced transition risk relative to Arm A.

Table 18: Multi-state transition hazard ratios by treatment arm

Treatment Arm	Stable to Progression HR (95% CI)	Progression to Death HR (95% CI)
Arm A (baseline)	1.00	1.00
Arm B	0.56 (0.35–0.88)	1.37 (0.81–2.31)
Arm C	0.59 (0.38–0.92)	1.02 (0.62–1.69)

The multi-state model indicated that Arms B and C reduced the transition hazard from stable disease to progression compared with Arm A. Arm B had a stable-to-progression hazard ratio of 0.56, while Arm C had a stable-to-progression hazard ratio of 0.59. There was no clear evidence that either arm changed the progression-to-death transition when treatment arm alone was included in the multi-state model.

The multi-state model presented above included treatment arm as the sole covariate, using the full available sample. To assess whether the treatment-arm effect was robust to adjustment for baseline prognostic factors, a second multi-state model was fitted including treatment arm together with number of metastatic sites, LDH level, TMB status, and JAK mutation status as covariates on both transition intensities. This adjusted model used the complete-case sample, reflecting the same reduction in sample size described for the logistic regression and Cox models in Sections 3.4 and 3.6.

Table 17 below summarises the key adjusted transition-specific hazard ratios from the complete-case multi-state model.

Table 19: Adjusted multi-state transition hazard ratios for treatment arms and baseline covariates

Covariate	Stable to Progression HR (95% CI)	Progression to Death HR (95% CI)
Arm B vs Arm A	0.66 (0.27–1.59)	4.91 (1.44–16.80)
Arm C vs Arm A	0.99 (0.46–2.16)	0.36 (0.14–0.97)
Sites >= 3	1.68 (0.84–3.37)	1.50 (0.61–3.69)
LDH Elevated	1.08 (0.52–2.24)	0.93 (0.37–2.36)
TMB >= 10	0.83 (0.40–1.75)	0.85 (0.29–2.45)
JAK Mutated	0.40 (0.16–1.01)	3.57 (1.29–9.91)

Comparing the adjusted model to the crude, treatment-arm-only model, the Arm B stable-to-progression effect weakened after adjustment (HR 0.56 to 0.66), and the Arm C effect moved from protective to approximately null (HR 0.59 to 0.99). Sites >= 3 and LDH elevation showed hazard ratios above 1 for stable-to-progression, while TMB >= 10 showed a slightly protective direction; however, their confidence intervals were wide and crossed 1.

The progression-to-death transition showed a more striking shift. After adjustment, Arm B's hazard of death following progression increased and became statistically significant (HR = 4.91, 95% CI 1.44–16.80), while Arm C's decreased and also became statistically significant (HR = 0.36, 95% CI 0.14–0.97). JAK mutation itself showed a large increase in post-progression mortality risk (HR = 3.57, 95% CI 1.29–9.91), despite showing a protective trend for progression. Sites, LDH, and TMB did not show clear evidence of association with the progression-to-death transition because their confidence intervals crossed 1.

These adjusted post-progression estimates should be treated cautiously. The wide confidence intervals, notably 1.44–16.80 for Arm B, reflect very few death events in the reduced complete-case sample, and the large reversal from the crude model likely reflects a combination of JAK-related confounding and instability from fitting a five-covariate model on a much smaller dataset. This result is therefore exploratory rather than confirmatory.

Tables 18 and 19 and Figures 7, 8, and 9 below show state occupation probabilities over time estimated from the multi-state model. These outputs show the predicted probabilities of remaining stable, progressing, or dying over follow-up.

Table 20: Twelve-month transition probabilities from stable disease

Arm	P(Stable at 12 months)	P(Progression at 12 months)	P(Death at 12 months)
A	0.728	0.147	0.125
B	0.838	0.074	0.088
C	0.830	0.092	0.078

Table 21: State occupation probabilities across time by treatment arm

Time	Stable_A	Stable_B	Stable_C	Death_A	Death_B	Death_C
0	1.00	1.00	1.00	0.00	0.00	0.00
2	0.95	0.97	0.97	0.01	0.00	0.00
4	0.90	0.94	0.94	0.02	0.01	0.01
6	0.85	0.92	0.91	0.04	0.03	0.02
8	0.81	0.89	0.88	0.06	0.05	0.04
10	0.77	0.86	0.86	0.09	0.07	0.06
12	0.73	0.84	0.83	0.12	0.09	0.08

Across time, Arms B and C maintained higher probabilities of remaining stable than Arm A. By 12 months, the probability of remaining stable was approximately 0.73 for Arm A, 0.84 for Arm B, and 0.83 for Arm C. The probability of death was also higher for Arm A than for Arms B and C at the same time point.

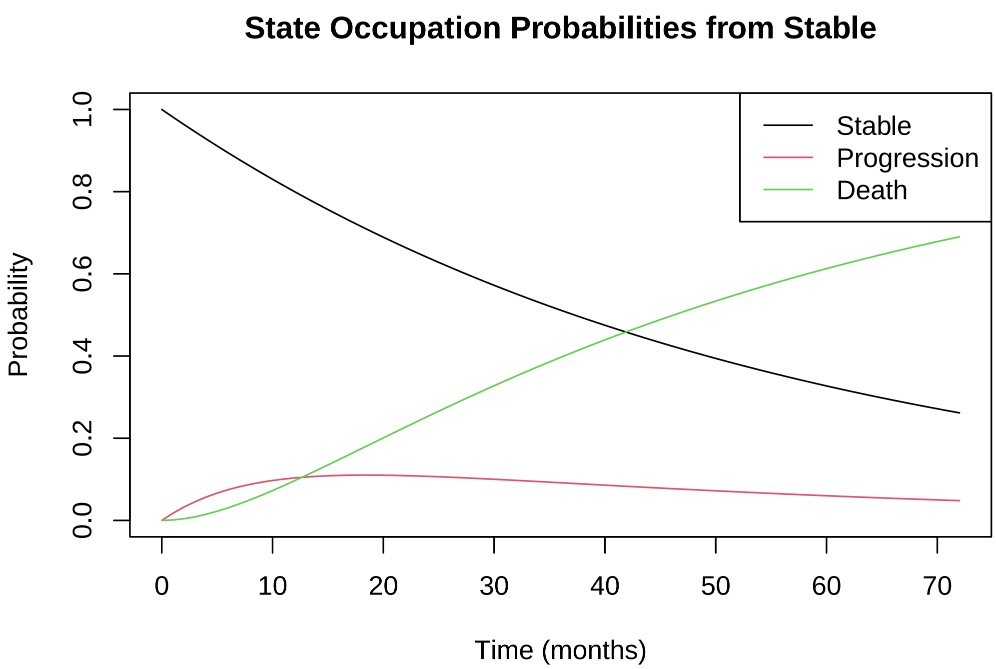

Figure 7: State occupation probabilities from stable disease

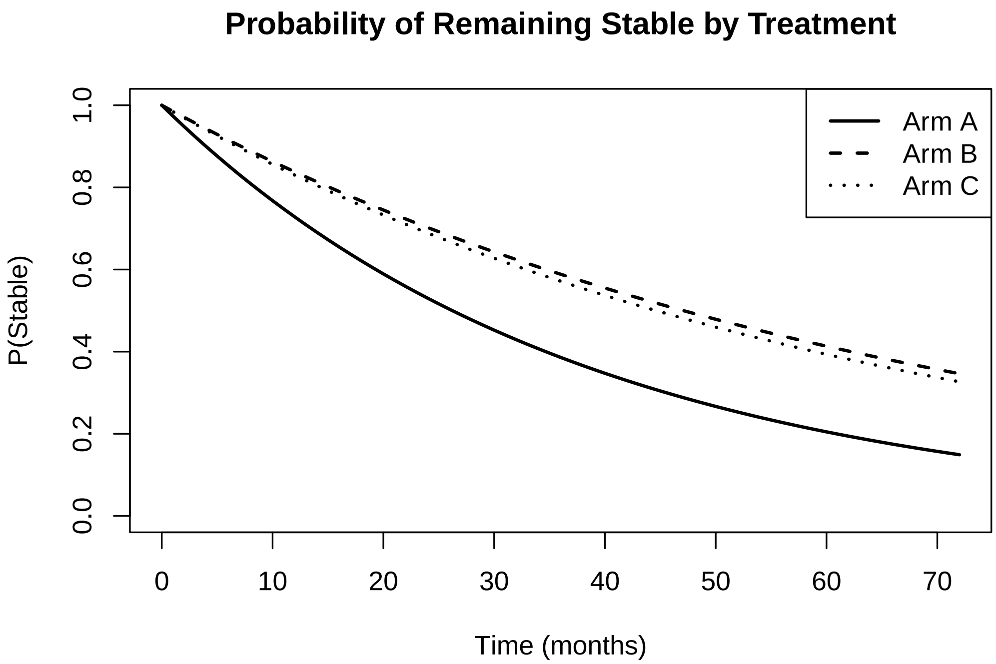

Figure 8: Probability of remaining stable by treatment arm

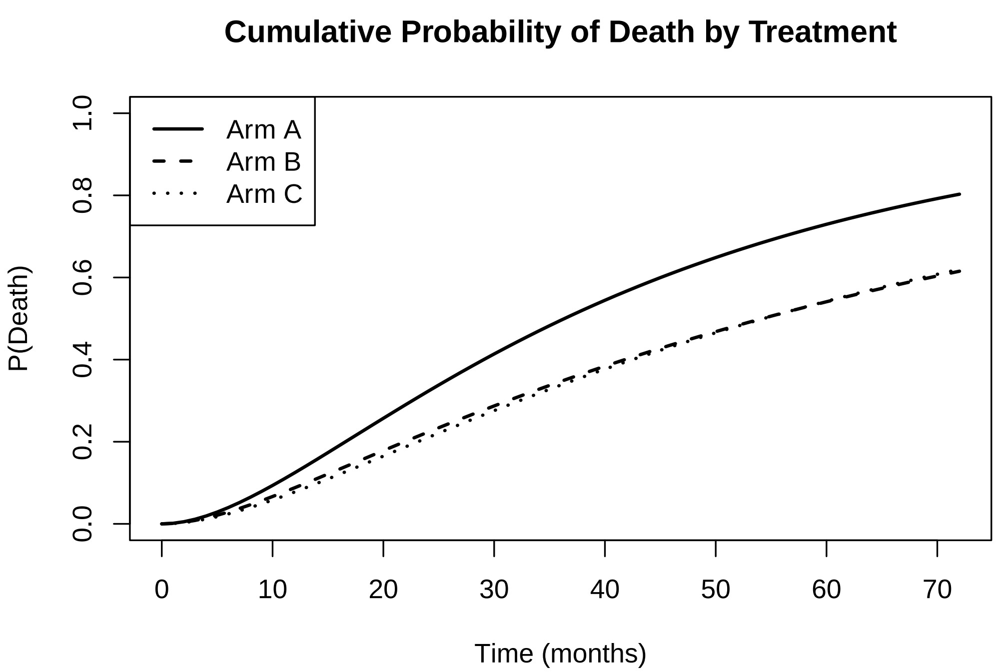

Figure 9: Cumulative probability of death by treatment arm

3.10 Comparison of Methods

Table 20 below provides a structured comparison of all statistical methods applied in this project across six criteria. Methods are ordered from least to most capable of handling multiple outcomes simultaneously.

Table 22: Comparison of statistical methods for multiple outcomes

Method	Multiple outcomes	Time-to-event	Censoring	Covariates	Output	Limitation
Logistic regression	Composite only	No	No	Yes	Odds ratio	Ignores timing
Kaplan–Meier	Composite only	Yes	Yes	No	Survival curve	No adjustment
Log-rank test	Composite only	Yes	Yes	No	p-value	No effect size
Cox model	Composite only	Yes	Yes	Yes	Hazard ratio	PH assumption
Win ratio	Yes	Yes	Yes	Limited	Win ratio	Needs hierarchy
Net benefit	Yes	Yes	Yes	Limited	Net benefit	Pairwise framework
Multi-state model	Yes	Yes	Yes	Yes	Transition probabilities	Transition assumptions

The comparison shows that logistic regression, Kaplan–Meier analysis, log-rank tests, and Cox models are useful but limited when outcomes are related and sequential. Win-ratio and net-benefit methods better incorporate clinical hierarchy, while multi-state models provide the most complete representation of disease pathways over time.

4 Discussion

4.1 Main Findings

This project compared several statistical strategies for handling multiple outcomes in health data. Applied to the SECOMBIT dataset specifically, OS did not show a statistically significant difference across treatment arms, while PFS showed stronger evidence of treatment-arm differences. The composite endpoint analysis was nearly identical to the PFS analysis because no patient died before progression. Win-ratio and net-benefit analyses suggested favourable trends for Arms B and C compared with Arm A. The multi-state model showed that Arms B and C mainly delayed progression from stable disease, rather than clearly changing mortality after progression.

4.2 Interpretation of Treatment Sequencing

The results suggest that treatment sequencing may influence disease progression dynamics. Arm A began with encorafenib plus binimetinib and switched to ipilimumab plus nivolumab after progression. Arm B began with ipilimumab plus nivolumab and switched to encorafenib plus binimetinib after progression. Arm C began with encorafenib plus binimetinib for eight weeks before switching to ipilimumab plus nivolumab until progression.

Table 20 below summarises the treatment sequencing strategies evaluated in the SECOMBIT trial.

Treatment Arm	Initial Treatment	Treatment Switch	Subsequent Treatment
Arm A	Encorafenib + Binimetinib	After disease progression	Ipilimumab + Nivolumab
Arm B	Ipilimumab + Nivolumab	After disease progression	Encorafenib + Binimetinib
Arm C	Encorafenib + Binimetinib (8-week induction)	Planned switch after 8 weeks	Ipilimumab + Nivolumab until progression, followed by Encorafenib + Binimetinib

Table 20. Treatment sequencing strategies evaluated in the SECOMBIT trial.

The multi-state model suggested lower stable-to-progression transition hazards for Arms B and C compared with Arm A. This may indicate that earlier exposure to immunotherapy is associated with delayed progression. However, these findings should be interpreted cautiously because the study had a relatively small sample size and was not powered to definitively test transition-specific treatment sequencing hypotheses.

4.3 Composite Endpoint and PFS Similarity

A key finding was that PFS and the composite endpoint produced almost identical results. This occurred because every progression event was also counted as a composite event, and no patient died before progression. Therefore, in this dataset, the composite endpoint was effectively driven by progression rather than death.

This is important because it shows that the similarity between PFS and composite results was not a coding error. It also highlights a limitation of composite endpoints: if one component dominates the composite, the result may not reflect the full clinical hierarchy of outcomes. In this case, the composite endpoint did not add substantial information beyond PFS.

4.4 Why the Methods Give Different Results

The differences in p-values and effect sizes across methods reflect their distinct mathematical structures rather than inconsistencies in the data. The log-rank test treats all events within a given endpoint as equivalent and asks whether the observed number of events in each arm differs from expectation under the null hypothesis. It therefore gives the greatest statistical weight to the most frequent event, which in this dataset is progression.

The win ratio uses a different principle. Every patient in one arm is compared with every patient in another arm, and pairs are resolved hierarchically. OS is evaluated first, and PFS is used only when OS is tied. This means the win ratio gives mathematical priority to death, even though death is less frequent than progression. As a result, the win-ratio p-values for B vs A and C vs A can appear more favourable than the OS log-rank p-value alone because the win ratio accumulates information from both OS and PFS in clinical order of importance.

The net-benefit results reflect the same hierarchical structure. The OS contribution to the global net benefit can be larger than the PFS contribution even when OS is not individually significant, because every OS win in a matched pair is counted before any PFS win. This comparison highlights the key limitation of the composite endpoint in this dataset: because no patient died before progressing, the composite endpoint was driven entirely by progression and did not give additional priority to death.

4.5 Novel Findings

The most clinically interesting finding was the possible effect of treatment sequencing on disease progression. The early Kaplan–Meier curves suggested a change in treatment-arm pattern around the two-month time point, which corresponds to the planned switch in Arm C. This supports the hypothesis that the timing of immunotherapy exposure may influence early disease control.

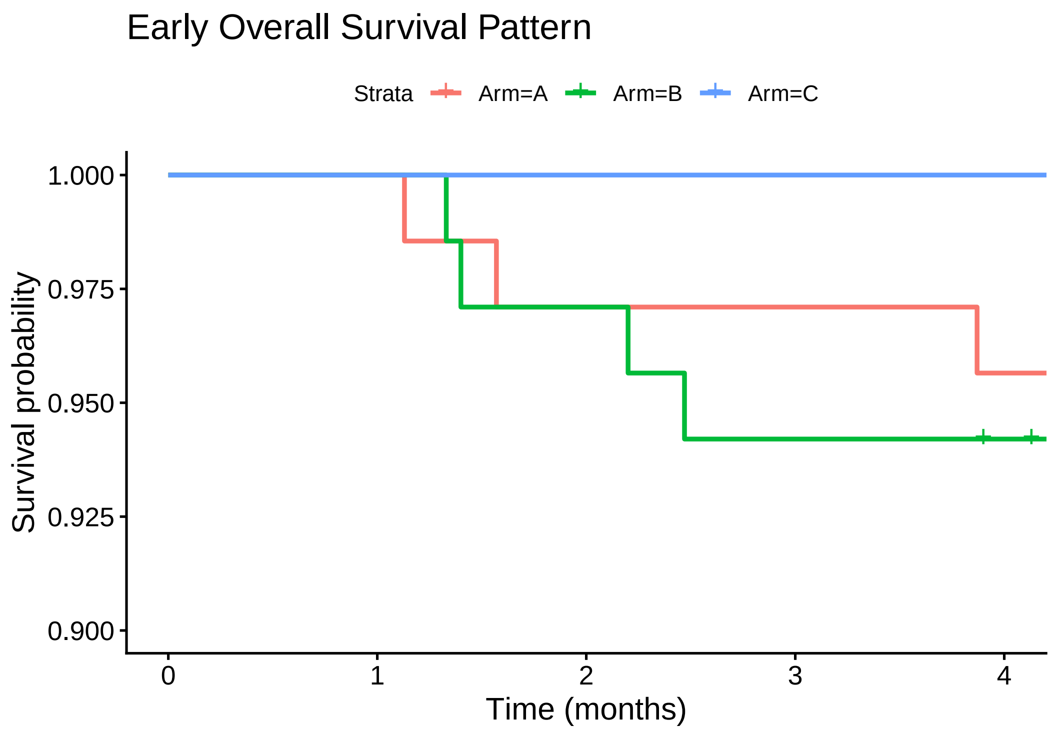

Figure 10: Early overall survival pattern up to four months

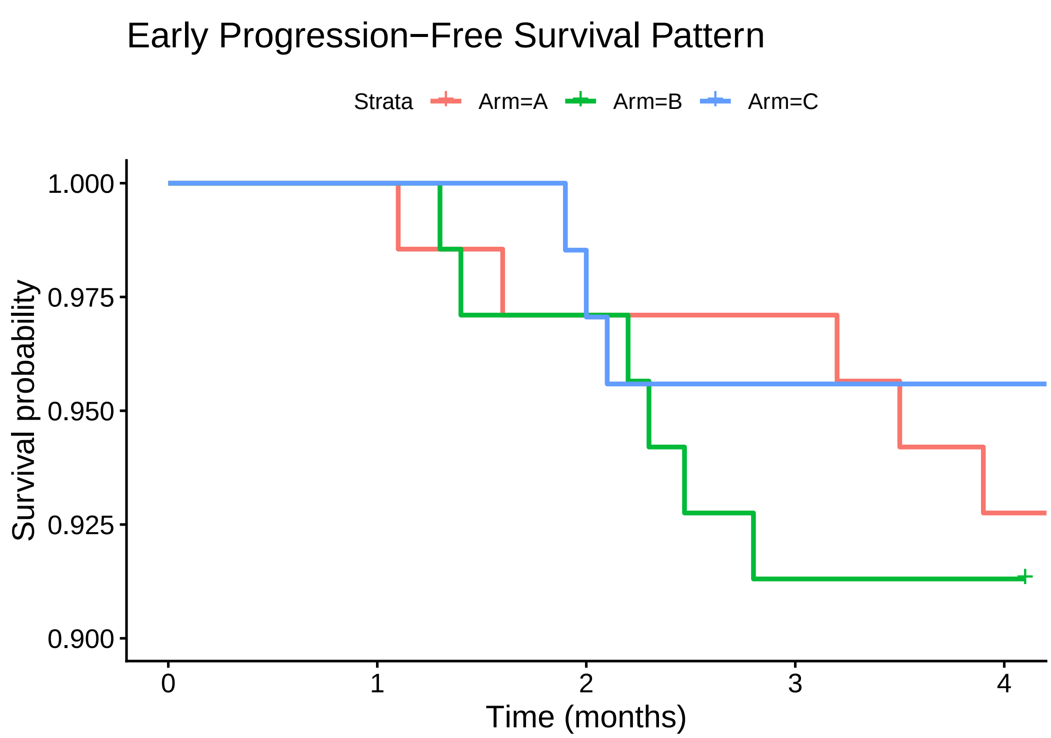

Figure 11: Early progression-free survival pattern up to four months

Inspection of the early curves suggests that the treatment-arm pattern may change around the two-month point. This is clinically important because Arm C switches treatment at approximately eight weeks. Although this finding is exploratory, it suggests that ipilimumab plus nivolumab may contribute to delayed progression when used early in the treatment sequence. This interpretation is consistent with the broader SECOMBIT findings that immunotherapy-first or planned-switch strategies may be favourable for many patients with metastatic BRAF V600-mutated melanoma (20,21).

4.5.1 Drug Swapping and Treatment Sequencing Strategy

A closer examination of the treatment administration schedule across the three arms helps explain the early Kaplan–Meier crossover described above. All three arms in SECOMBIT eventually received both treatment classes: targeted therapy (encorafenib plus binimetinib) and immunotherapy (ipilimumab plus nivolumab). However, the order and trigger for swapping between them differed substantially by arm.

Arm	First-line treatment	Switch trigger	Second-line treatment
A	Encorafenib plus binimetinib	On progression (reactive)	Ipilimumab plus nivolumab after progression
B	Ipilimumab plus nivolumab	On progression (reactive)	Encorafenib plus binimetinib after progression
C	Encorafenib plus binimetinib for fixed eight-week induction	Planned switch (proactive)	Ipilimumab plus nivolumab until progression, then encorafenib plus binimetinib

In Arm A, encorafenib plus binimetinib was given first, and the swap to ipilimumab plus nivolumab occurred only after disease progression. In Arm B, this sequence was reversed: ipilimumab plus nivolumab was given first, with the swap to encorafenib plus binimetinib occurring only after progression. In Arm C, encorafenib plus binimetinib was given for a fixed induction period of eight weeks, after which patients were switched to ipilimumab plus nivolumab on a pre-planned schedule, regardless of whether progression had occurred, with the swap back to encorafenib plus binimetinib reserved for after progression on immunotherapy.

This distinction separates the three arms into two different sequencing ideas. Arms A and B used reactive second-line switches, where the drug class was changed only once progression had already been confirmed. Arm C's first switch was proactive, because the swap to immunotherapy was scheduled in advance and did not wait for evidence of treatment failure.

The early-curve analysis in Section 4.5, Figures 10 and 11, showed a crossover at approximately the two-month time point across both the OS and PFS panels. This timing corresponds to the point at which Arm C executed its planned switch from targeted therapy to immunotherapy. Before this point, Arm C tracked closely with Arm A; shortly afterward, Arm C's trajectory shifted, while Arm B, which had been receiving immunotherapy from treatment initiation, showed the clearest separation from the other two arms. Taken together with the multi-state model results in Section 3.9, this suggests that earlier exposure to immunotherapy, introduced either immediately or on a fixed schedule rather than in response to progression, was associated with better early disease control than withholding immunotherapy until first-line treatment had already failed.

This observation has a direct practical implication for treatment sequencing strategy. Arm A's design, which delayed immunotherapy until progression on targeted therapy was confirmed, was associated with the least favourable outcomes across several endpoints examined in this project, including the highest event counts (Table 4), less favourable win-ratio and net-benefit comparisons (Tables 14 and 15), and the highest stable-to-progression transition hazard (Table 16). By the time progression is detected under a reactive strategy, the disease has typically already advanced past the point at which first-line therapy was providing adequate control, and switching to a second treatment class at that stage may come too late to fully recapture the benefit that earlier exposure might have provided.

By contrast, both arms that introduced immunotherapy before confirmed progression, Arm B immediately and Arm C on a fixed eight-week schedule, showed more favourable disease trajectories. This suggests that clinicians and trial designers should be cautious about using disease progression as the sole trigger for swapping drug classes. A planned or earlier introduction of immunotherapy, rather than a swap reserved exclusively for treatment failure, appeared to produce better early and sustained disease control in this dataset.

This finding should be interpreted cautiously. The early-curve crossover is based on a small number of events within the first four months of follow-up, and the trial was not formally powered to test the timing of the swap as an isolated variable independent of the overall treatment sequence. Nonetheless, the consistency between the early Kaplan–Meier crossover, the multi-state transition hazards, and the win-ratio and net-benefit comparisons gives reasonable support to the conclusion that avoiding a purely progression-triggered drug swap, in favour of an earlier or scheduled introduction of immunotherapy, was associated with improved outcomes in this analysis.

This raises a further question: is switching drug classes necessary at all, or could continuing immunotherapy alone, without ever introducing targeted therapy, match or outperform the sequencing strategies tested here? Arm B still switches to encorafenib plus binimetinib once progression occurs, so it does not isolate immunotherapy as a standalone strategy. Since SECOMBIT did not include an immunotherapy-only arm, the dataset cannot establish whether the later switch adds any benefit beyond what early immunotherapy exposure already provides. A future trial arm using continued immunotherapy with no planned switch would help determine whether Arm B's advantage stems from the early use of immunotherapy itself, or from the specific sequence in which the two drug classes are administered.

4.6 Strengths

A strength of this project is that it compared traditional and advanced statistical methods using the same clinical dataset. This allowed direct comparison across methods in terms of censoring, timing, covariate adjustment, clinical hierarchy, and disease pathway interpretation. Another strength is the use of multi-state modelling, which provided clinically interpretable transition probabilities rather than only endpoint-specific hazard ratios.

4.7 Limitations

The main limitation was the relatively small sample size, with approximately 68–69 patients in each arm. This reduced statistical power, especially for pairwise treatment comparisons and transition-specific multi-state modelling. Missingness in the JAK mutation variable reduced the effective sample size for adjusted models. The proportional hazards assumption was also questionable for some endpoints, which limits interpretation of Cox model hazard ratios.

Another limitation was that the dataset contained only one observed transition pathway: stable disease to progression and progression to death. Since no patient died before progression, the multi-state model could not estimate a direct stable-to-death transition. This limits generalisability to datasets where death before progression is observed.

4.8 Clinical Trial and Predictive Modelling Implications

In this project, the methods were mainly used for clinical trial inference: estimating whether average outcomes differed across treatment arms. The randomised SECOMBIT design supports this use because treatment arm was assigned by design rather than chosen by clinicians or patients.

The multi-state model also has predictive modelling potential. The state occupation probabilities estimate the probability that a patient starting in stable disease will remain stable, progress, or die at future time points. If individual-level covariates such as LDH, TMB, and JAK mutation status are added to the transition models, this framework could generate patient-specific probability estimates. This is a major advantage over single-endpoint Cox models, log-rank tests, win ratios, and net-benefit summaries, which mainly describe average group-level treatment differences.

4.9 Future Work

Future research should apply these methods to larger datasets with more complex disease pathways and greater numbers of outcome events. Larger datasets would improve statistical power for transition-specific modelling and would allow direct stable-to-death transitions, competing risks, recurrent events, and time-dependent covariates to be explored.

Future work could also develop dynamic prediction models that estimate an individual patient's probability of remaining stable, progressing, or dying at future time points. This would make multi-state models more directly useful for clinical decision-making. Additional studies are also needed to investigate the potential treatment sequencing effects suggested by the SECOMBIT analysis, particularly the role of early immunotherapy exposure.

5 Conclusion

Traditional survival methods such as Kaplan–Meier analysis, log-rank tests, and Cox models are useful for analysing individual time-to-event outcomes. However, they are limited when outcomes are related, hierarchical, or sequential. Composite endpoints provide a simple combined outcome but may hide important differences between clinically unequal components. Logistic regression is especially limited for this type of data because it ignores event timing and censoring.

The win-ratio and net-benefit methods improved interpretation by incorporating clinical hierarchy and overall treatment advantage. The multi-state model provided the most complete view of disease progression because it described how patients moved from stable disease to progression and death over time.

In this SECOMBIT analysis, Arms B and C appeared to delay progression compared with Arm A, while there was no clear evidence of a difference in post-progression mortality. The multi-state model was able to show this transition-specific pattern directly: by separating the disease pathway into a stable-to-progression transition and a progression-to-death transition, it revealed that the benefit of Arms B and C was concentrated in delaying progression rather than in extending survival after progression, a distinction that single-endpoint methods could not have made. This pointed toward early exposure to immunotherapy, rather than the act of switching itself, as the more likely driver of improved outcomes. The findings suggest that advanced methods, particularly multi-state models, can provide more clinically meaningful insight than analysing multiple outcomes separately.

6 Project Link

The code repository for this project is available at: https://github.com/darshu-d/MSc-Research-project-

References

1. Follmann D. Multivariate tests for multiple endpoints in clinical trials. Statistics in Medicine. 1995;14(11):1163–76.

2. Neaton JD, Gray G, Zuckerman BD, Konstam MA. Key issues in end point selection for heart failure trials: Composite end points. Journal of Cardiac Failure. 2005;11(8):567–75.

3. Freemantle N, Calvert M, Wood J, Eastaugh J, Griffin C. Composite outcomes in randomized trials: Greater precision but with greater uncertainty? JAMA. 2003;289(19):2554–9.

4. Kaplan EL, Meier P. Nonparametric estimation from incomplete observations. Journal of the American Statistical Association. 1958;53(282):457–81.

5. Mantel N. Evaluation of survival data and two new rank order statistics arising in its consideration. Cancer Chemotherapy Reports. 1966;50:163–70.

6. Cox DR. Regression models and life-tables. Journal of the Royal Statistical Society: Series B. 1972;34(2):187–220.

7. Kleinbaum DG, Klein M. Survival analysis: A self-learning text. Springer; 2012.

8. Pocock SJ, Ariti CA, Collier TJ, Wang D. The win ratio: A new approach to the analysis of composite endpoints in clinical trials based on clinical priorities. European Heart Journal. 2012;33(2):176–82.

9. Rauch G, Kieser M, Binder H, Bayes-Genis A, Jahn-Eimermacher A. Time-to-first-event versus recurrent-event analysis in clinical trials. Statistics in Medicine. 2014;33(13):2163–77.

10. Unverzagt S et al. Composite endpoints in clinical trials. Trials. 2019.

11. Cordoba G, Schwartz L, Woloshin S, Bae H, Gotzsche PC. Definition, reporting, and interpretation of composite outcomes in clinical trials: Systematic review. BMJ. 2010;341:c3920.

12. Therneau TM, Grambsch PM. Modeling survival data: Extending the cox model. Springer; 2000.

13. Kalbfleisch JD, Prentice RL. The statistical analysis of failure time data. Wiley; 2002.

14. Andersen PK, Borgan O, Gill RD, Keiding N. Statistical models based on counting processes. Springer; 1993.

15. Vickers AJ, Van Calster B, Steyerberg EW. Net benefit approaches to the evaluation of prediction models, molecular markers, and diagnostic tests. BMJ. 2016;352:i6.

16. Peron J, Roy P, Ozenne B, Roche L, Buyse M. The net chance of a longer survival as a patient-oriented measure of treatment benefit in randomized clinical trials. JAMA Oncology. 2016;2(7):901–5.

17. Jackson CH. Multi-state models for panel data: The msm package for r. Journal of Statistical Software. 2011;38(8):1–29.

18. Putter H, Fiocco M, Geskus RB. Tutorial in biostatistics: Competing risks and multi-state models. Statistics in Medicine. 2007;26(11):2389–430.

19. Wreede LC de, Fiocco M, Putter H. The mstate package for estimation and prediction in non- and semi-parametric multi-state and competing risks models. Computer Methods and Programs in Biomedicine. 2010;99(3):261–74.

20. Ascierto PA et al. Sequencing of ipilimumab plus nivolumab and encorafenib plus binimetinib for untreated BRAF-mutated metastatic melanoma (SECOMBIT): A randomized, three-arm, open-label phase II trial. Journal of Clinical Oncology. 2023.

21. Ascierto PA et al. Sequential immunotherapy and targeted therapy for metastatic BRAF V600 mutated melanoma: 4-year survival and biomarkers evaluation from the phase II SECOMBIT trial. Nature Communications. 2024;15:146.

22. Little RJA, Rubin DB. Statistical analysis with missing data. Wiley; 2019.

23. Buuren S van. Flexible imputation of missing data. CRC Press; 2018.

24. Hosmer DW, Lemeshow S, Sturdivant RX. Applied logistic regression. Wiley; 2013.

25. Grambsch PM, Therneau TM. Proportional hazards tests and diagnostics based on weighted residuals. Biometrika. 1994;81(3):515–26.

26. Buyse M. Generalized pairwise comparisons of prioritized outcomes in the two-sample problem. Statistics in Medicine. 2010;29(30):3245–57.

27. Therneau TM. Survival: Survival analysis. 2024.

28. Kassambara A, Kosinski M, Biecek P. Survminer: Drawing survival curves using ggplot2. 2021.

29. Ozenne B, Buyse M. BuyseTest: Generalized pairwise comparisons. 2024.

30. R Core Team. R: A language and environment for statistical computing. R Foundation for Statistical Computing; 2024.

31. Royston P, Parmar MKB. Restricted mean survival time: An alternative to the hazard ratio for the design and analysis of randomized trials with a time-to-event outcome. BMC Medical Research Methodology. 2013;13:152. doi:10.1186/1471-2288-13-152

32. Harrington DP, Fleming TR. A class of rank test procedures for censored survival data. Biometrika. 1982;69(3):553–66. doi:10.1093/biomet/69.3.553

33. Fleming TR, Harrington DP. Counting processes and survival analysis. New York: John Wiley & Sons; 1991.

34. Anderson JR, Cain KC, Gelber RD. Analysis of survival by tumor response. Journal of Clinical Oncology. 1983;1(11):710–9. doi:10.1200/JCO.1983.1.11.710